



MUTUAL FUND ANALYTICS

Final Project Report

Technologies: Python, Pandas, NumPy, SQL, SQLite, SciPy, Plotly, Seaborn, Power BI, Git/GitHub

Dataset Coverage: 40 mutual-fund schemes and 10 core datasets

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1. Executive Summary

The **Mutual Fund Analytics Project** is an end-to-end Data Analyst project designed to analyze mutual-fund industry trends, fund performance, investor behavior, portfolio concentration and market-related indicators.

The project combines multiple structured datasets containing information about mutual-fund schemes, daily NAV, assets under management, SIP inflows, category inflows, investor transactions, portfolio holdings and benchmark indices.

The project follows a complete analytics workflow:

Data Collection → Data Cleaning → Data Validation → Database Design → SQL Analysis → EDA → Performance Analytics → Advanced Analytics → Power BI Dashboard → Final Reporting

2. Project Introduction

Mutual funds generate large volumes of financial and investor data. This data can be used to understand how funds perform, how investors behave and how the overall mutual-fund industry changes over time.

However, raw datasets are often stored separately and may contain:

- Different date formats
- Missing values
- Duplicate records
- Different categorical values

- Numeric fields stored as strings
- Different levels of aggregation
- Multiple benchmark series

Therefore, before performing analysis, the data must be cleaned and transformed into an analytical structure.

This project addresses this problem by creating a complete data analytics pipeline.

3. PROBLEM STATEMENT

The mutual fund datasets used in this project originate from multiple sources and have different structures, formats, and business meanings.

Without proper processing, it is difficult to:

- Compare funds consistently.
- Analyze historical NAV movements.
- Measure risk-adjusted performance.
- Track AUM growth.
- Understand SIP trends.
- Analyze investor behavior.
- Identify concentrated portfolios.
- Compare funds with market benchmarks.
- Present results to business users.

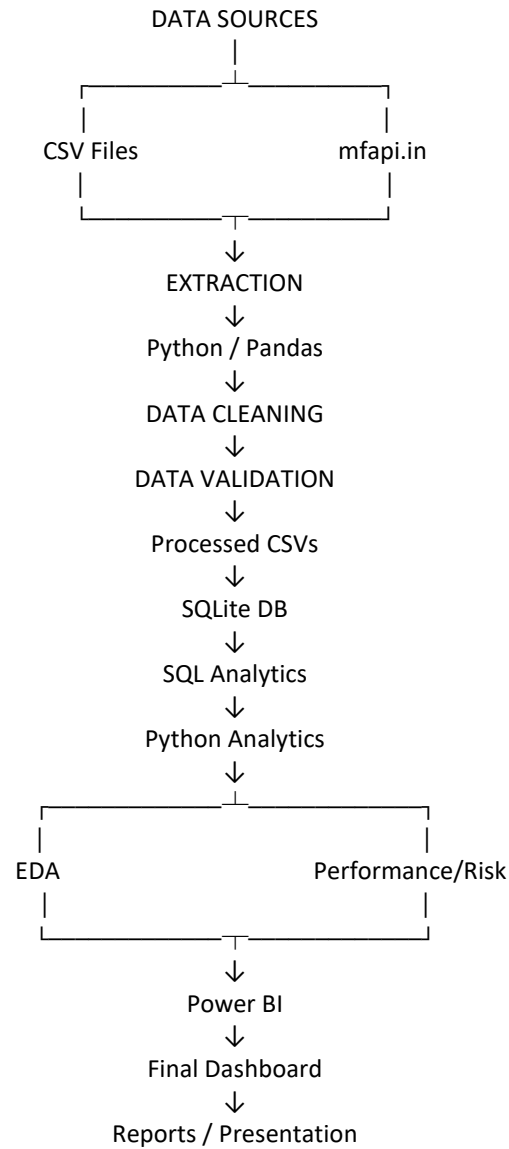
4. DATA SOURCES

The project uses 10 major datasets.

No.	Dataset	Purpose
1	fund_master.csv	Fund information
2	nav_history.csv	Historical NAV
3	aum_by_fund_house.csv	AUM by fund house
4	monthly_sip_inflows.csv	SIP activity
5	category_inflows.csv	Category inflows
6	industry_folio_count.csv	Folio statistics
7	scheme_performance.csv	Fund performance
8	investor_transactions.csv	Investor transactions
9	portfolio_holdings.csv	Portfolio allocation
10	benchmark_indices.csv	Benchmark data

5. SYSTEM ARCHITECTURE

The overall architecture follows an ETL and analytics pipeline.



6.DATA INGESTION AND ETL

The ETL process consists of three major stages.

Extract

Data was extracted from:

- CSV datasets
- mfapi.in API

Python's Pandas library was used to load CSV files.

The API response was retrieved using Python's requests library.

Transform

Data transformation included:

- Date conversion
 - Sorting
 - Duplicate removal
 - Missing-value handling
 - Numeric conversion
 - Validation of transaction types
 - Validation of AMFI codes
 - Validation of NAV values
 - Expense ratio validation
-

Load

Cleaned datasets were stored in:

data/processed/

They were then loaded into SQLite using SQLAlchemy.

7.DATA CLEANING AND VALIDATION

NAV Cleaning

The following operations were performed:

1. Convert date to datetime.
2. Sort by AMFI code and date.
3. Remove duplicate records.
4. Validate NAV values.
5. Ensure NAV > 0.
6. Handle missing observations where appropriate.

Daily returns were calculated using:

$$\text{Daily Return} = (\text{NAV}_t / \text{NAV}_{t-1}) - 1$$

Investor Transaction Cleaning

Transaction types were standardized to:

SIP

Lumpsum

Redemption

Transaction amounts were validated to ensure:

amount > 0

Dates were standardized and KYC status values were checked.

Scheme Performance Cleaning

Return fields were converted to numeric format.

The following fields were validated:

- 1-year return

- 3-year return
- 5-year return
- Benchmark return
- Alpha
- Beta
- Sharpe
- Sortino
- Standard deviation
- Maximum drawdown

Expense ratios were checked against a reasonable range.

AMFI Validation

AMFI codes in fund_master were compared against AMFI codes in nav_history.

The purpose was to ensure that every fund referenced in the master dataset had corresponding historical NAV data.

8. DATABASE DESIGN

A SQLite database named:

bluestock_mf.db

was created.

The database follows a structured analytical/star-schema approach.

Dimension tables

- dim_fund
- dim_date

Fact tables

- fact_nav
- fact_transactions
- fact_performance
- fact_aum

SQL ANALYTICS

SQL was used to answer business questions.

Examples include:

Top 5 funds by AUM

Identifies funds with the highest assets under management.

Average NAV by month

Used to understand monthly NAV behavior.

SIP YoY growth

Used to measure annual changes in SIP inflows.

Transactions by state

Used to identify geographic investment patterns.

Funds with expense ratio below 1%

Used to identify relatively low-cost schemes.

9.EXPLORATORY DATA ANALYSIS

EDA was performed using:

- Matplotlib

- Seaborn
- Plotly

The analysis covered the period **2022–2026**, depending on dataset availability.

10.1 NAV Trend Analysis

Historical NAV trends were plotted for the 40 schemes.

The chart helps identify:

- Long-term growth
- Short-term volatility
- Market corrections
- Differences between schemes

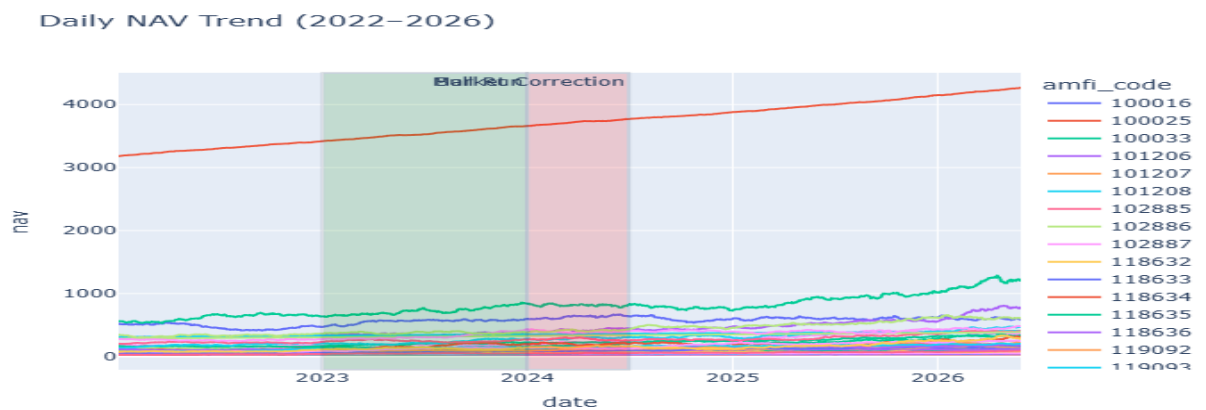
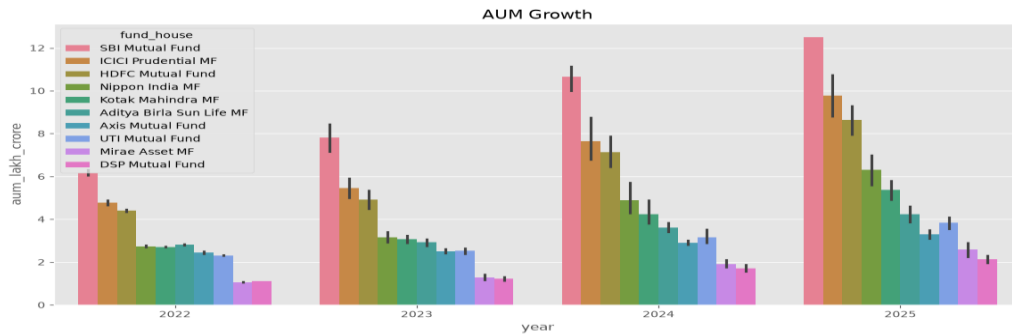


Figure 1 – NAV Trend of 40 Mutual Fund Schemes

9.2 AUM Growth

AUM was analyzed by fund house and year.

The analysis identifies major fund houses by assets under management and shows changes over time.



9.3 SIP Inflow Trend

Monthly SIP inflows were analyzed from January 2022 through December 2025.

The analysis highlights the growth of systematic investment activity.

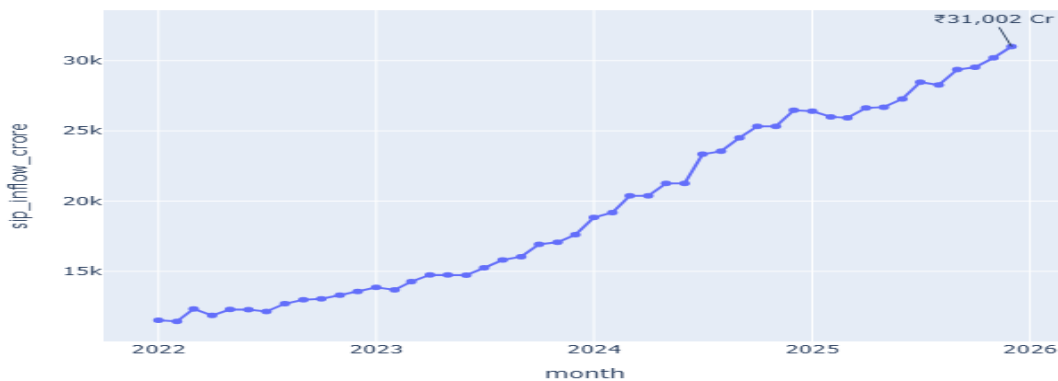


Figure 3 – Monthly SIP Inflow Trend

9.4 Category Inflow Heatmap

A heatmap was created to compare monthly net inflows across fund categories.

The visualization helps identify:

- Strong inflow periods
- Weak periods

- Categories attracting investors
- Seasonal patterns

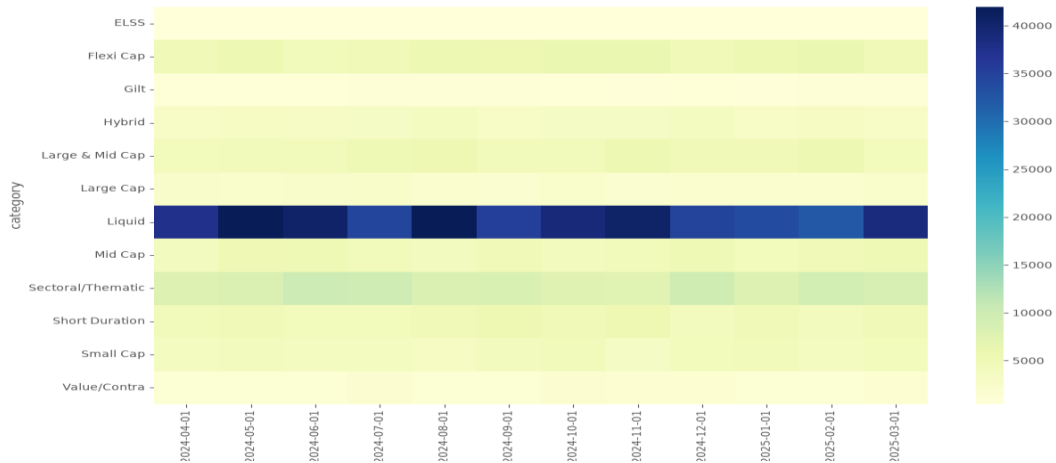


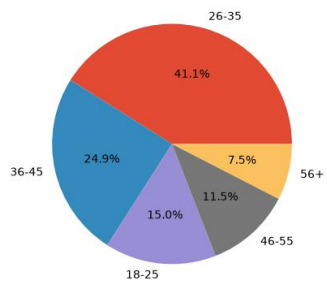
Figure 4 – Category Inflow Heatmap

9.5 Investor Demographics

Investor transactions were analyzed by:

- Age group
- Gender
- State
- City tier

Pie and bar charts were used to visualize investor distribution.



9.6 Geographic Distribution

Transaction amounts were aggregated by state.

This helps identify regions contributing significantly to investment activity.

City-tier analysis compared:

- T30
- B30

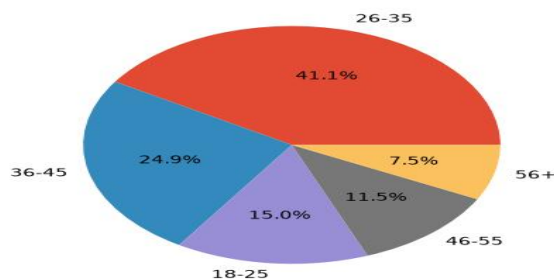


Figure 4 – Demographics Pie

9.7 Folio Growth

Industry folio counts were analyzed over time.

The project uses the provided industry figures, including:

- **13.26 Cr** at the beginning of the referenced period
- **26.12 Cr** at the end of the referenced period

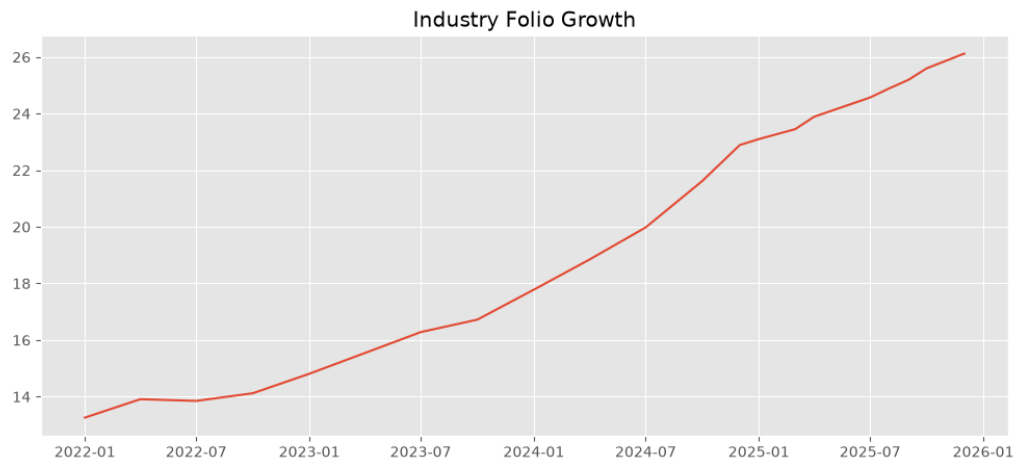


Figure 5 – Folio Count Growth

9.8 Sector Allocation

Portfolio holdings were aggregated by sector.

A donut chart was created to show the relative allocation of assets across sectors.

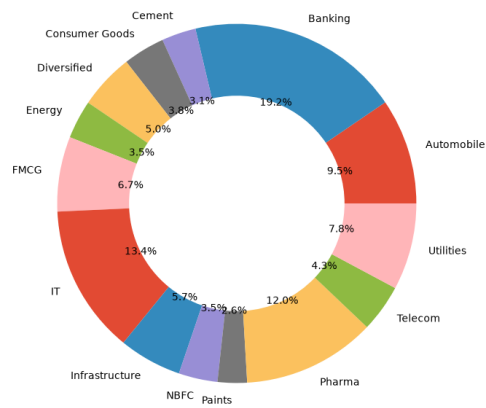


Figure 6 – Sector Allocation

10. POWER BI DASHBOARD

The Power BI dashboard consists of four pages.

Page 1 — Industry Overview

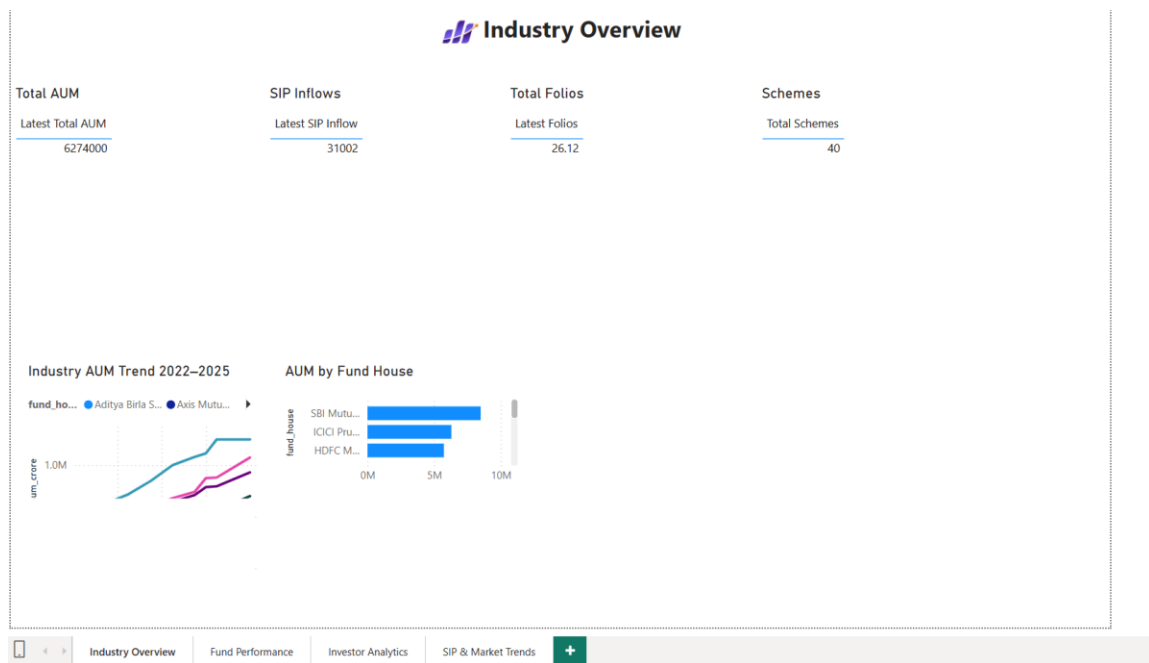
The Industry Overview page contains:

KPI Cards

- Total AUM
- SIP Inflows
- Folios
- Number of Schemes

Charts

- Industry AUM trend
- AUM by AMC



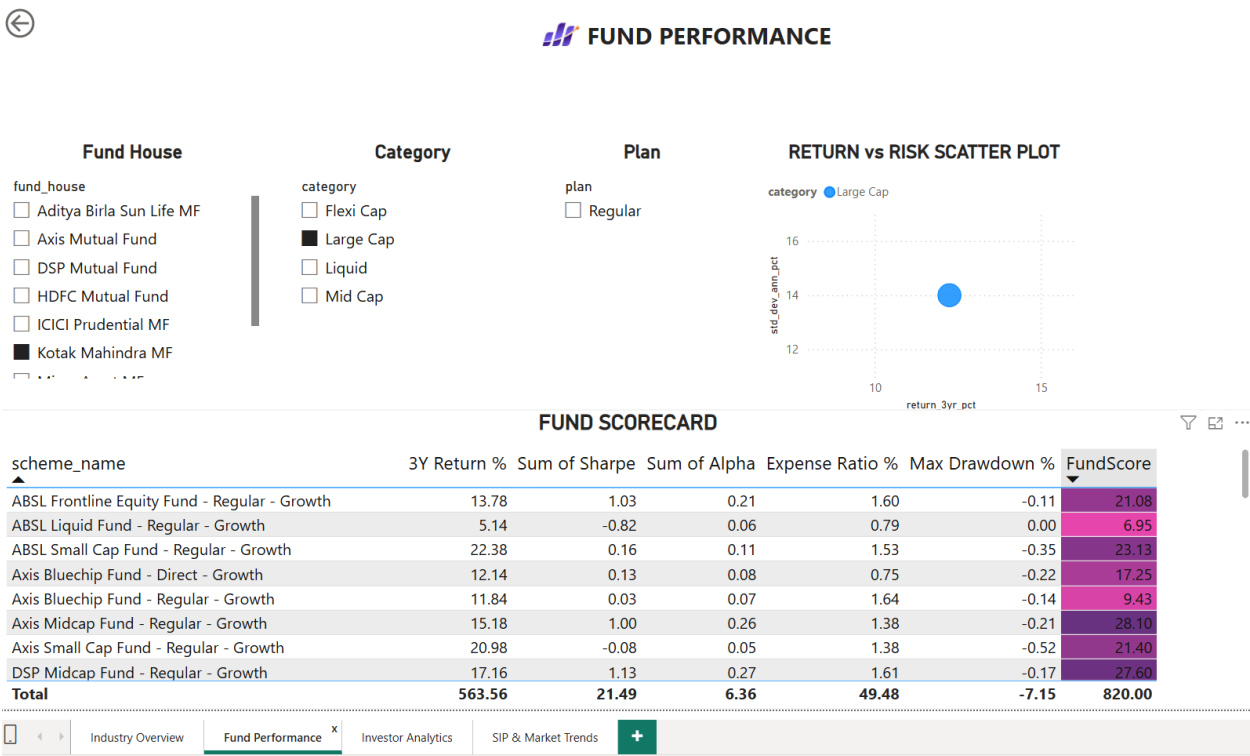
Page-1 : Industry Overview

Page 2 — Fund Performance

This page contains:

- Return vs Risk scatter plot
- Fund scorecard table
- NAV vs benchmark
- Fund House slicer
- Category slicer
- Plan slicer

The page allows users to compare funds interactively.

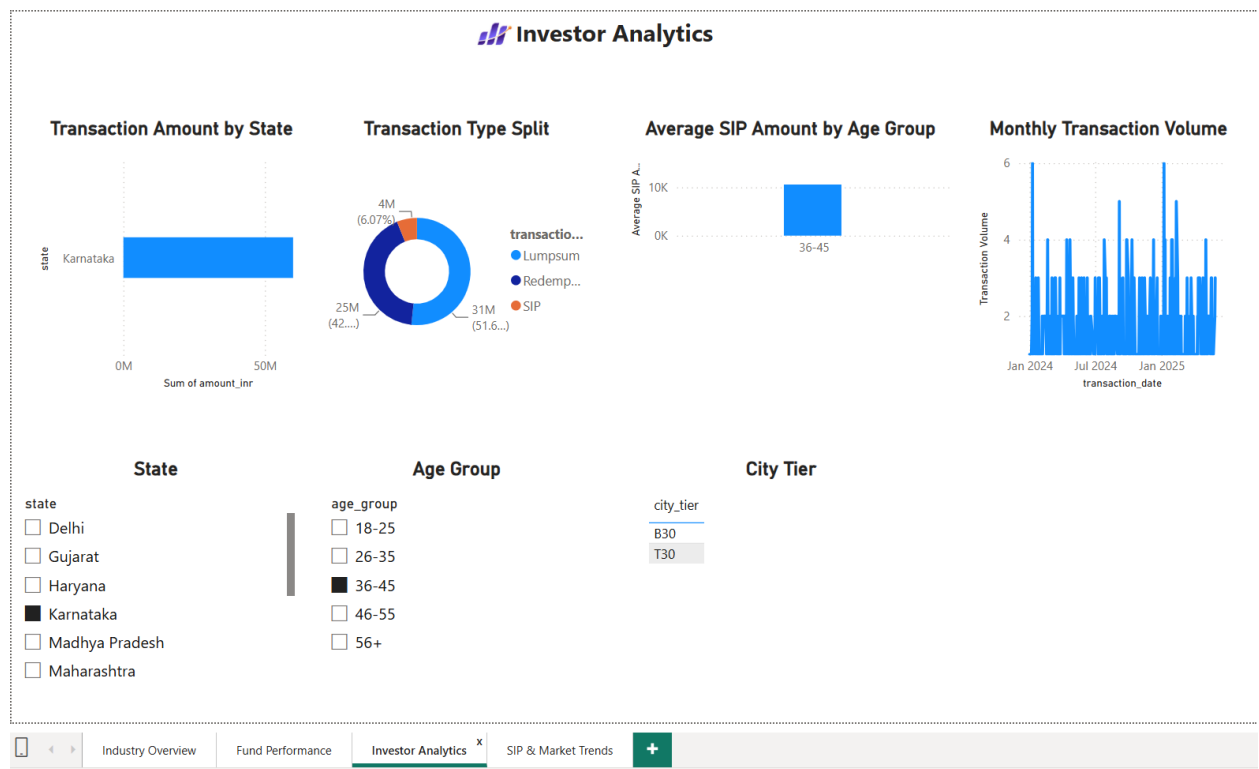


Page 2 : Fund Performance

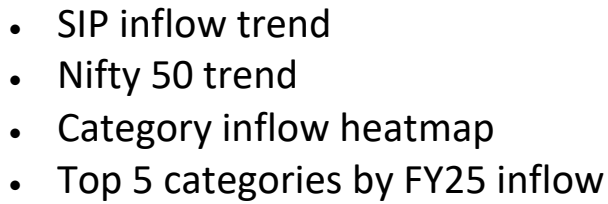
Page 3 — Investor Analytics

Visualizations include:

- Transaction amount by state
- Transaction type donut
- Age group vs average SIP amount
- Monthly transaction volume
- State slicer
- Age group slicer
- City tier slicer



Page 3: Investor Analytics



12. LIMITATIONS

The project has several limitations.

1. Historical Data

Historical performance does not guarantee future performance.

2. Risk-Free Rate

A fixed risk-free-rate proxy of 6.5% was used for the requested analysis. Actual risk-free rates vary over time.

3. Historical VaR

Historical VaR assumes that the historical return distribution provides useful information about future risk.

4. Dataset Coverage

The analysis depends on the coverage and quality of the provided datasets.

5. Benchmark Differences

Different mutual fund schemes may have different appropriate benchmarks.

6. Investor Data

Investor transaction data may not represent the complete Indian mutual fund investor population.

7. Recommendation System

The recommender is a simplified analytical model based primarily on risk grade and Sharpe Ratio. It is not a substitute for professional financial advice.

8. Data Quality

Any errors or inconsistencies present in source datasets may affect analytical results.